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Asymmetric Petasis reaction for the synthesis of chiral α- and β-butadienyl amines†

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The Petasis reaction using (1*S*,2*R*)-1-amino-2-indanol as the substrate and an activator to construct α- and β-butadienyl amines in optically pure forms was realized, which are otherwise difficult to prepare. The reactions feature a metal-free nature, broad substrate scope, complete regioselectivities (γ-selectivity of pinacol homoallenyl- and isoprenylboronates), and high to excellent chirality induction (up to >20 : 1 dr). The favored nucleophilic addition across the Si-face of the imine intermediate was explained using DFT calculations of the six-membered chair-like transition state.

Petasis reaction[1] using three components of carbonyls, amines and boron-reagents, as the nucleophile-variation of classic Mannich reaction, has become a versatile tool for the construction of enantiopure functionalized amine derivatives.[2] Apart from the diol,[3] hydroxy–thiourea[4] and transition-metal[5] catalysis, neighbouring hydroxyl group activation through boronate ligand exchange between a hydroxyl group and a boronate ester or acid reagent has proved to be a robust and reliable strategy for the nucleophilic addition to imine intermediates *via* a closed six-membered transition structure.[6] This features a metal-free and easily-handled transformation, as well as a three-component one-pot synthesis without the requirement of isolating imines in advance. By employing this tactic, the hydroxyl group at the α-position of aldehyde or amine components enables the synthesis of α-functional amines from various boron-containing reagents, such as aryl,[7] allenyl and alkenyl[8] and allyl reagents.[9] Yet the application of this synthetic strategy with novel allylboron-type reagents to address the synthetic difficulty of some important chiral scaffolds remains rather underexplored, albeit highly in demand.

α-Butadienyl amines (2-aminomethyl-1,3-dienes) are valuable synthons for diarylmethylamines[10] through Diels–Alder reactions. They are also important structural units in natural products, such as dienic α-amino acids found in the leaves of the ornamental shrub *Philadelphus coronaries* (Fig. 1).[11] Thus far, some synthetic strategies have been documented to build this type of molecules. On the contrary to the gradual progress in racemic synthesis,[12] the corresponding asymmetric version remains rare. In 2005, Botta's group reported the synthesis of chiral 2-(*N*-1-acetyl-1-arylmethyl)-1,3-butadienes through the microwave-assisted ethylene–alkyne cross-metathesis of ethylene and chiral 1-arylpropargyl amides[13] (Scheme 1a).[14] Hiemstra's group demonstrated the synthesis of enantiopure diene amino acids *via* the addition of allenylmethylsilane into *N*-acyliminium ion, followed by an enzymatic kinetic resolution (Scheme 1b).[15] In this context, the development of a facile and general asymmetric strategy for the construction of chiral α-butadienyl amine derivatives remains a challenging, but rewarding task.

Since the Petasis reaction process is a γ-selective substitution of allylboronates on the basis of the Zimmerman–Traxler model, no rearrangement in the carbon chain of β-substituted allylboronates (*e.g.* isoprenylboronates) will occur when transforming into the final products; however, the carbon chain of γ-substituted allylboronates (*e.g.* homoallenylboronates) will be changed. As it turns out, the structural modification at various sites of allylboronates entails important change in the backbone of products. Previously, we have applied pinacol homoallenyl- and isoprenylboronates into the installation of chiral 1,3-butadienyl[16] and isoprenyl[17] carbinols through a Brønsted acid activation strategy, respectively. They both feature the γ-regioselectivity of two boronate reagents. We envisage that these two boronate-reagents could be utilized in the synthesis

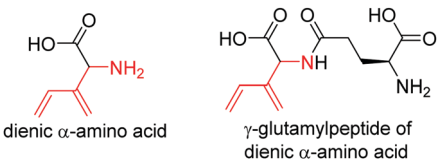


Fig. 1 Chiral α-butadienyl amine natural products.

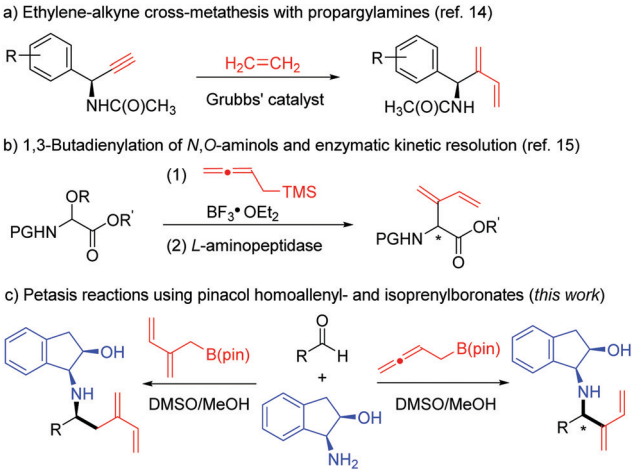
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† Electronic supplementary information (ESI) available. CCDC 2023204. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0cc08241d

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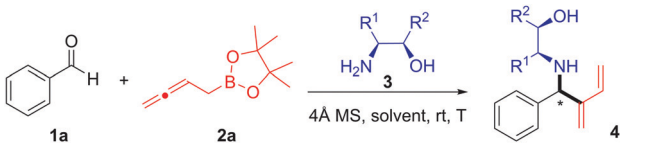
Scheme 1 Various synthetic strategies for installing chiral 1,3-dienylamines.

of chiral α-butadienyl amines and β-butadienyl amines by virtue of an asymmetric Petasis reaction (Scheme 1c). As a part of our continuous efforts in developing novel synthetic methodologies with boronate reagents,[18] we herein report this study.

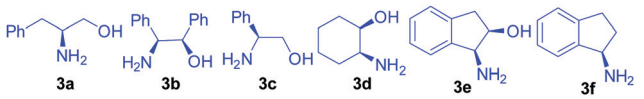
To validate the feasibility of the asymmetric Petasis reaction using aldehyde **1a** and pinacol homoallenylboronate **2a** for the synthesis of chiral 2-α-butadienyl amines **4**, we commenced our study on screening various chiral 1,2-aminols **3**, since the hydroxy group is essential for the Petasis reaction.[6] The key task is to solve the diastereoselective problem. In the presence of 50.0 mg of a molecular sieve (4 Å MS) in MeOH at room temperature (r.t.), chiral 1,2-aminols **3a–c** were investigated, delivering chiral α-butadienyl amines **4** in good yields but with low diastereoselectivities (Table 1, entries 1–3). No reaction using (1*S*,2*R*)-2-aminocyclohexanol **3d** was found (entry 4). (1*S*,2*R*)-1-Amino-2-indanol **3e** displayed better performance, affording the corresponding product in 94% yield with 3 : 1 dr within 12 h (entry 5). To our surprise, the reaction without any molecular sieves proceeded much faster, which suggests that a small amount of H_2O generated *in situ* from the condensation of amine and aldehyde is probably beneficial to releasing product *via* hydrolysis (4 h *vs.* 12 h, entries 5 and 6). In order to improve the diastereoselective result with **3e**, different combinations of a non-protic solvent and 5.0 equiv. of MeOH were studied (entries 7–11; see Table S1 in the ESI†); to our delight, the DMSO media enabled an enhanced dr (11.3 : 1) and a satisfactory yield (95%, entry 11; see Table S1 in ESI†), which was identified as the optimal reaction conditions. For comparison, indanamine **3f** was used under otherwise identical reaction conditions, or at an elevated temperature; however, the reaction did not proceed at all; this indicates the essential role of a hydroxy group (entry 12).

Under the optimized reaction conditions, a number of aryl and aliphatic aldehydes were readily scoped (Scheme 2). Firstly, the effects of different substituents (*R*′) at the phenyl group on the reactivity and diastereoselectivity were evaluated. The incorporation of electron-donating groups (Me– and MeO–) at the *para*-position resulted in inferior dr values (**4b** and **4c**);

Table 1 Optimized conditions[a]



Entry	Aminol 3	Solvent	4A MS mg^{-1}	t h^{-1}	Yield[b] (%)	dr[c]
1	**3a**	MeOH	50.0	48	68	2 : 1
2	**3b**	MeOH	50.0	48	65	1.3 : 1
3	**3c**	MeOH	50.0	72	86	2 : 1
4	**3d**	MeOH	50.0	72	—[d]	—
5	**3e**	MeOH	50.0	12	94	3 : 1
6	**3e**	MeOH	—	4	95	3.4 : 1
7	**3e**	THF + 5.0 equiv. MeOH	—	48	52	3 : 1
8	**3e**	Toluene + 5.0 equiv. MeOH	—	48	72	3 : 1
9	**3e**	CH_2Cl_2 + 5.0 equiv. MeOH	—	48	75	3.5 : 1
10	**3e**	EtOAc + 5.0 equiv. MeOH	—	72	87	3.6 : 1
11	**3e**	DMSO + 5.0 equiv. MeOH	—	20	95	11.3 : 1
12	**3f**	DMSO + 5.0 equiv. MeOH	—	20	—[d]	—

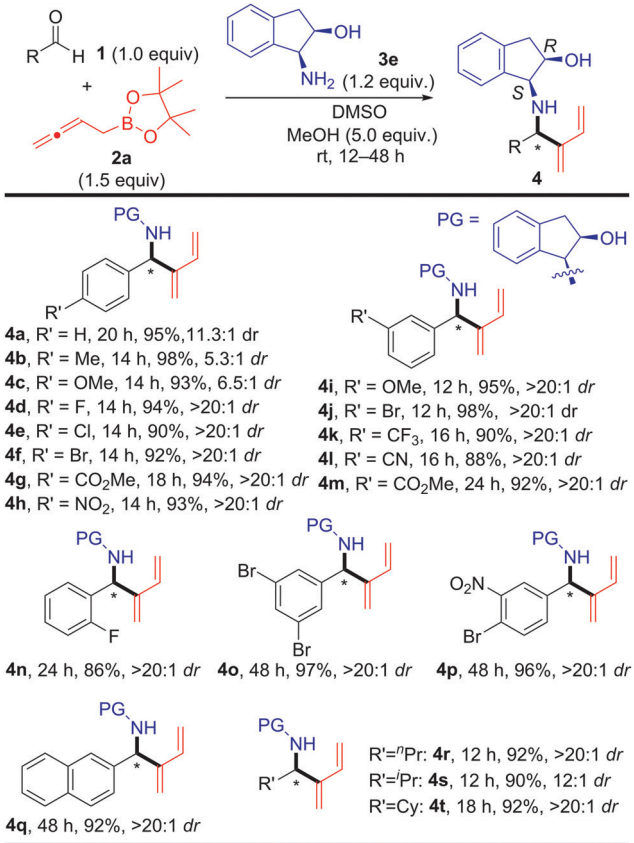


[a] **1** (0.2 mmol), **3** (0.24 mmol), and solvent (0.4 mL) were stirred at r.t. for 2 h, then **2a** (0.30 mmol) was added. [b] Isolated yield. [c] Diastereomeric ratio (dr) was determined by ^{1}H NMR. [d] No reaction.

however, the electron-neutral (F–, Cl– and Br–) and electron-withdrawing (–CO_2Me and –NO_2) groups were more beneficial to the diastereoselective outcome. In these cases, 90% to 94% yields and all >20 : 1 dr were observed (**4d–h**). For the *meta*-substituted aldehydes, perfect dr values were obtained independent of the electronic nature of the substituents; whereas, the yields of the examples with electron-withdrawing substituents (–CF_3, –CN and –CO_2Me; **4k–m**) were slightly lower than those of the other two cases (**4i, j**). A single example of *ortho*-substituted phenyl aldehyde was studied to provide product **4n** in 86% yield with >20 : 1 dr. In addition, two disubstituted aromatic aldehydes were also tolerated in the Petasis reaction with excellent yields and dr values (**4o, p**). 2-Naphthylaldehyde underwent Petasis 1,3-butadienylation smoothly to give product **4q** in 92% yield with >20 : 1 dr. Finally, three aliphatic aldehydes were successfully employed in the reaction, affording products **4r–t** in high yields with perfect dr values except isobutyraldehyde **4s** (12 : 1 dr).

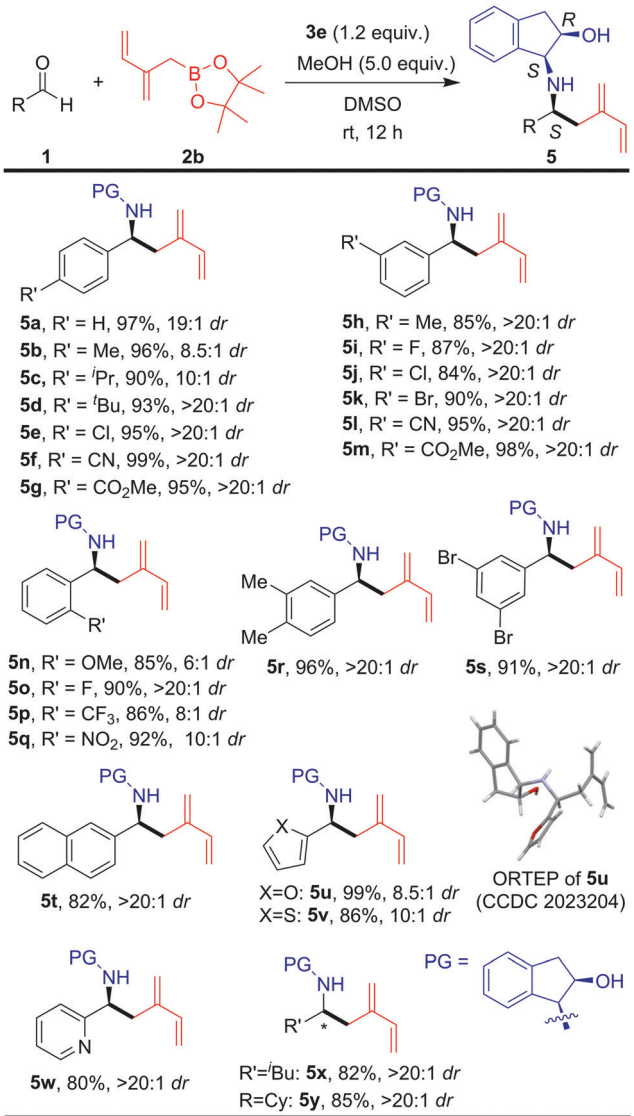
β-Butadienyl amine (2-aminoethyl-1,3-diene or isoprenylamine)[19] is the homologue of α-butadienyl amine, which can be prepared *via* a similar synthetic strategy–Petasis reaction with pinacol isoprenylboronate **2b**. Compared with the reaction conditions mentioned above for Petasis butadienylation, less volume of DMSO (0.2 mL) was required for the Petasis isoprenylation to promote the complete consumption of the aldehyde starting material. As depicted in Scheme 3, a wide range of aromatic aldehydes with varied

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[a] **1** (0.2 mmol), **3e** (0.24 mmol) and MeOH (5.0 equiv.) were stirred in DMSO (0.4 mL) at rt for 2 h, then **2a** (0.30 mmol) was added; Isolated yield; Diastereomeric ratio (*dr*) was determined by ^{1}H NMR.

Scheme 2 Scope of aldehydes with boronate **2a**[a].



[a] **1** (0.2 mmol), **3e** (0.24 mmol) and MeOH (5.0 equiv.) were stirred in DMSO (0.2 mL) at rt for 2 h, then **2b** (0.30 mmol) was added.

Scheme 3 Scope of aldehydes with boronate **2b**[a].

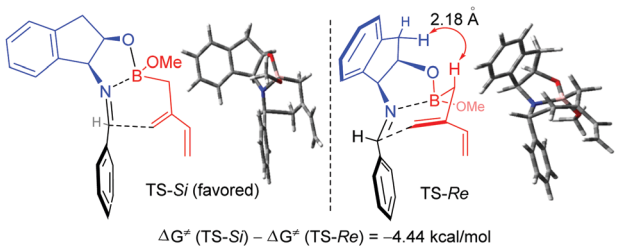


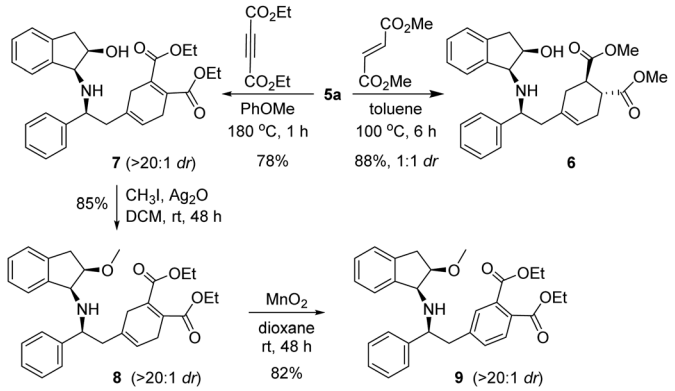
Fig. 2 Proposed transition states.

substituents and aliphatic aldehydes participated in the Petasis isoprenylation reaction. Monosubstituted phenyl aldehydes including electron-rich and electron-poor substituents (*R′*) at the *ortho*-, *meta*- and *para*-positions were first studied, providing the corresponding β-butadienyl amines **5b–q** in 84–99% yields with up to >20 : 1 dr. Electron-rich groups (except ^{t}Bu-group) at the *para*-position and substituents at the *ortho*-position (except F-atom) were not good for the diastereoselectivity, and dr values lower than 20 : 1 were obtained. In addition, 3,4-dimethyl and 3,5-dibromo phenyl aldehydes were also applied to furnish products **5r, s** in high yields with perfect dr values. The use of 2-naphthyl and heteroaromatic aldehydes afforded products **5t–w** in 80–99% yields with 8.5 : 1 to >20 : 1 dr; the absolute configuration of the newly formed stereogenic carbon center of **5u** was unambiguously assigned to be *S* based on the results of X-ray crystallographic analysis.[20] Thus, the absolute configurations of other aryl isoprenylated products were determined by analogy. Finally, two aliphatic aldehydes were used to produce β-butadienyl amines **5x** and **5y** in 82–85% yields with >20 : 1 dr.

In order to understand the observed stereochemical outcomes, density-functional theory (DFT) calculations of the key transition state have been carried out by using the Gaussian 09 suite of programs (see the details in the ESI†). As depicted in Fig. 2, the difference of calculated free energy barrier ($\Delta(\Delta G^{\ddagger})$ = −4.44 kcal mol^{-1}) between transition states TS-Si and TS-Re for the isoprenylboration reaction indicates that the process is favorable for addition through the Si-face of the imine part to generate the desired adduct **5a.** Based on the 3D model of the transition states, the indan moiety in TS-Re

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Scheme 4 Synthetic utility of isoprenylamine **5a**.

exhibits a much closer distance to the chair-like six-membered cycle (the nearest neighbor distance of 2.18 Å) than TS-Si, which may result in increased steric repulsion and a higher free energy barrier.

Toward exhibiting the synthetic utility of the isoprenylamine **5a**, Diels–Alder reactions using dimethyl fumarate and diethyl butynedioate to provide chiral cyclohexenyl (**6**, 88% yield and 1 : 1 dr) and cyclohexadienyl methylamines (**7**, 78% yield and >20 : 1 dr) were accomplished, respectively (Scheme 4). With compound **7** in hand, the asymmetric synthesis of optically pure 1,2-diarylethylamine **9** was completed through two steps of methylation and oxidative aromatization.

In summary, a synthetic approach to chiral α- and β-butadienyl amine skeletons has been established *via* asymmetric Petasis reactions with pinacol homoallenyl- and isoprenylboronates, respectively. A wide range of aromatic and aliphatic aldehydes were applied with up to 99% yield and >20 : 1 dr. (1*S*,2*R*)-1-amino-2-indanol plays a key role in enabling the transformation and chirality induction. The DFT calculations of the six-membered chair-like transition states shed light on the experimentally observed isoprenylboration across the Si-face of the imine intermediate. The synthetic application of isoprenylamine for chiral 1,2-diarylethylamine has been demonstrated. Further endeavours to achieve asymmetric catalysis with various allylboronates are ongoing in our lab.

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Conflicts of interest

There are no conflicts to declare.

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20 CCDC 2023204† contains the supplementary crystallographic data (**5u**) for this paper.